

PARTS LIST

Semiconductors:

IC1	- PIC16F877A microcontroller
IC2	- 7805 voltage regulator
IC3	- 7806 voltage regulator
LED1	- 5mm Light-emitting diode
D1	- 1N4148 diode
D2-D6	- 1N4007 diode
T1	- SL100 transistor

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon unless stated otherwise):

R1-R8, R10	- 4.7-kilo-ohm
R9	- 150-ohm
R11	- 10-kilo-ohm
R12	- 470-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2, C3	- 0.1 μ F ceramic
C4, C5	- 22pF ceramic

Miscellaneous:

X _{AL}	- 4MHz crystal oscillator
X1	- 230V AC primary to 0-9V, 500mA secondary transformer
RL1	- HD44780-based 16 $\times$ 2 LCD
SL-S17	- 6V, 1C/O relay
	- Push-to-on tactile switch

bit flash program memory, 368 bytes of RAM and many other internal peripherals like analogue-to-digital converter, USART, timers, synchronous serial port, compare captures and pulse-width modulation modules, EEPROM and analogue comparators.

The job of the MCU in this project is to receive signals from the input device (keypad) and take corresponding actions. Whenever any key is pressed on the keypad, the software program in the MCU identifies the pressed key and accordingly turns on or turns off the appliance. Simultaneously, it also displays a message on the LCD screen.

4 $\times$ 4 matrix keypad. A 4 $\times$ 4 matrix keypad is used to give commands and the password to the MCU. It consists of 16 keys (S2-S17) arranged in the form of a square matrix of four rows and four columns. Each key in the matrix is labeled according to the operation assigned to it. The connections from the pin-outs of the keypad to the MCU pins are shown in Fig. 1. Rows 1 through 4 are connected to pins RB3, RB2, RB1 and RB0 of Port B of the MCU, respectively. Columns 1 through

Functions of Various Keys of the Keypad and Their Labeling

S.No.	Row	Column	Label	Operation/digit entry
1	1	1	1	Digit '1'
2	1	2	2	Digit '2'
3	1	3	3	Digit '3'
4	1	4	Nil	No operation allotted
5	2	1	4	Digit '4'
6	2	2	5	Digit '5'
7	2	3	6	Digit '6'
8	2	4	Lock	Lock or turn off the appliance
9	3	1	7	Digit '7'
10	3	2	8	Digit '8'
11	3	3	9	Digit '9'
12	3	4	New	Change the password
13	4	1	C	Clear or backspace
14	4	2	0	Digit '0'
15	4	3	Go	Should be pressed after entering the password
16	4	4	Open	Open the lock (asks for password when pressed)

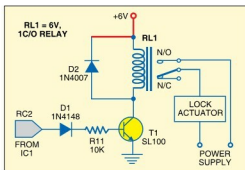


Fig. 3: Relay driver circuit

4 are connected to pins RB4 through RB7 of Port B, respectively.

16 $\times$ 2 LCD. A Hitachi HD44780 16 $\times$ 2 LCD is used to display various messages. It also displays an asterisk mark (*) for each digit of the password entered. Control lines EN, RW and RS of the LCD module are connected to pins RA1, RA2 and RA3 of Port A of the MCU, respectively. Commands and the data to be displayed are sent to the LCD module in nibble mode from Port D of the MCU. The higher four data bits of the LCD (D4 through D7) are connected to the lower nibble of Port D (RD0 through RD3) of the MCU.

Relay driver. RC2 pin of Port C of the MCU is interfaced with the relay driver circuit (shown in Fig. 3) to switch on or switch off the AC load (appliance). A relay driver circuit is nothing but a simple electronic circuit that drives an electromechanical relay. In this project, a 6V, single-changeover relay is used for switching the appliance 'on' or 'off.' Transistor SL100 plays the role of the relay driver.

Whenever the user enters the correct password, RC2 pin goes high (RC2=1). Consequently, transistor SL100 is triggered to energise the relay and the appliance turns 'on.' When RC2 is low

(RC2=0), the appliance turns 'off.' Free-wheeling diode 1N4007 protects the relay driver circuit from the reverse voltage developed in the relay coil.

You can also use optocoupler MCT2E to isolate the relay driver circuit from the microcontroller circuit. Whenever the user enters the correct password, RC2 pin goes high (RC2=1) and the internal LED of the MCT2E IC glows, which, in turn, triggers the internal transistor of MCT2E.

Power supply. Fig. 2 shows the power supply circuit. The 230V AC mains supply is stepped down to 9V AC using step-down transformer X1. The output from the secondary of the transformer is rectified by a bridge rectifier comprising diodes D3 through D6 and filtered by capacitor C1. The filtered output is regulated by ICs 7805 and 7806 connected in parallel to obtain the required 5V and 6V, respectively.

Software

The software code is written in 'C' language and compiled using Hitech C